FEDERAL TRADE COMMISSION 2026 · PERSONALIZED PRICING · 1,566 PUBLIC COMMENTS

Policy and Artificial Intelligence

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INFO 4340 Final Project



Human coding · machine learning · generative artificial intelligence · governance

THE OVERALL QUESTION

Can artificial intelligence help an agency read the public record?

Can it help make sense of thousands of public comments - accurately, affordably, and responsibly?

ACCURATE

Checked against independent human judgment

AFFORDABLE

Model cost measured for every run

GOVERNABLE

Bound by explicit operating rules

The point is not simply whether a model can classify text. The point is whether we can build a defensible workflow for a public decision context.



OUR CHOSEN CASE

Personalized pricing

A consumer-protection and public-policy classification problem

SPECIFIC QUESTION

What response does each commenter want from the Federal Trade Commission?

DOMAIN

Public policy · consumer protection · regulatory comment analysis

The Federal Trade Commission proposal focused on disclosure and enforcement against unfair or deceptive personalized-pricing practices; it did not propose a blanket ban.



Federal Trade Commission proposed enforcement policy statement, August 19, 2026

This project brings the course together

We moved from understanding the decision to building, evaluating, and governing a system.

UNDERSTAND

- Data audit and reproducibility
- Human coding and codebook design
- Inter-rater reliability

BUILD & COMPARE

- Traditional machine-learning baseline
- Prompt engineering and structured output
- Arena.ai: exploratory model comparison

EVALUATE

- Accuracy, precision and recall
- Macro-averaged F1 score
- Bootstrap, stability and cost

GOVERN

- Risk matrix and thresholds
- Human review and escalation
- Subgroup monitoring and accountability

Arena.ai was exploratory only. Formal performance evidence came from the reconciled human-coded validation set.

We measured policy intent, not sentiment

SENTIMENT

"Is this comment positive or negative?"

Too blunt. It can collapse very different regulatory positions into the same negative attitude toward personalized pricing.

POLICY INTENT

"What response does the commenter want from the Federal Trade Commission?"

- Stronger action
- Support as proposed
- Limit or oppose intervention
- Mixed or conditional
- Unclear

This framing became the backbone of our codebook, model prompts, and final policy analysis.

We built the pipeline in stages

The model was never our source of truth; reconciled human labels were the benchmark.

1

Collect & audit

1,566 individual text comments

2

Human code

40-case pilot + 120-comment validation sample

3

Establish reliability

85.0% agreement; Cohen's kappa 0.706

4

Compare models

Traditional baseline + Prompt A + Prompt B

5

Scale with guardrails

Full corpus, review queue, stability and cost

Key principle: we defined the coding rules before formal model evaluation and kept the pilot outside the validation sample.

Human coding gave us a defensible benchmark

We tested our own categories before asking a model to reproduce them.

85.0%

Raw agreement between our independent stance labels

0.706

Cohen’s kappa - above our 0.60 course threshold

120

Comments coded independently by both of us

What reconciliation changed

Most disagreements sat at the boundary between “unclear” and “stronger action.” We froze the rule that disliking personalized pricing is not automatically a request for government action; explicit requests to ban, stop, protect, opt out, or otherwise strengthen the remedy count as stronger action.

Prompt B gave the most balanced stance performance

Prompt A was a concise taxonomy. Prompt B encoded the frozen human codebook, boundary rules, and tie-breakers.

Macro-averaged F1 score



PROMPT B

80.8%

raw stance accuracy

93.5%

accuracy among validation comments not flagged for review

Why B? Prompt A had slightly higher raw accuracy, but Prompt B improved class-balanced performance and captured 78.3% of its stance errors with the review flag.

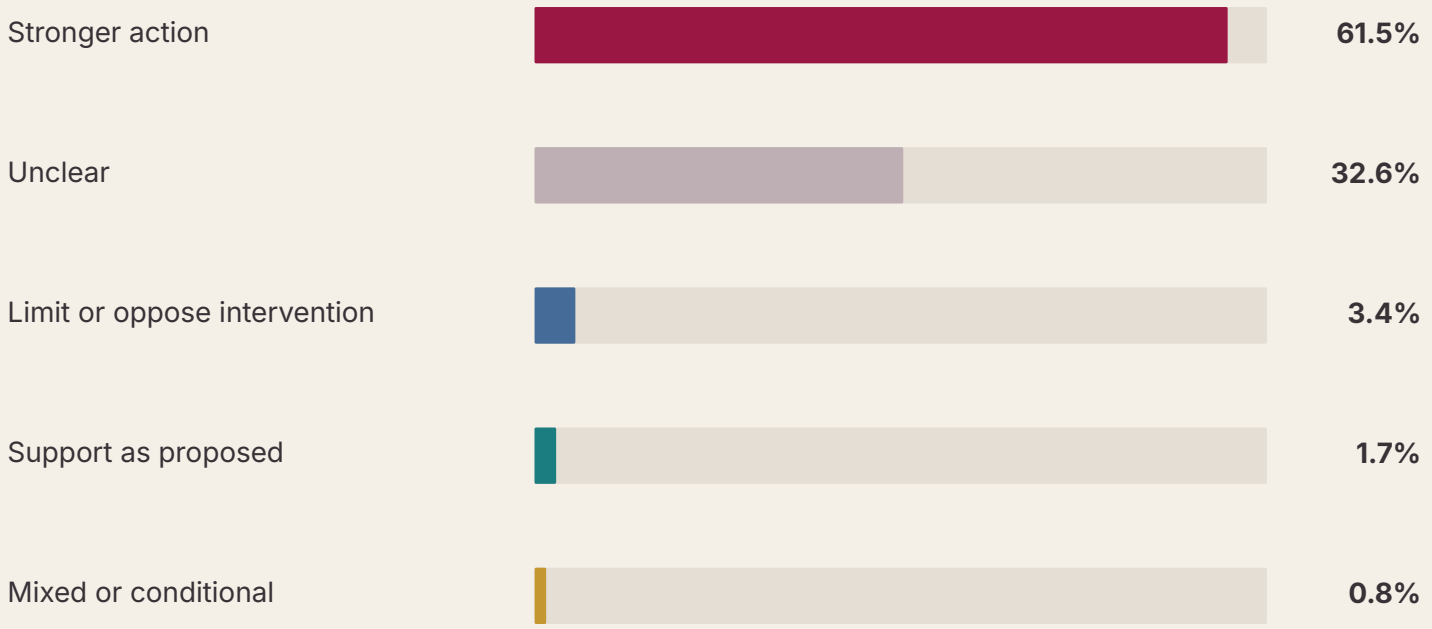
Arena helped us compare model behavior during development, but it was not part of the formal benchmark or model-selection evidence.

The model-coded corpus was dominated by calls to go further

Prompt B classified all 1,566 individual text comments using the frozen stance definitions.

61.5%

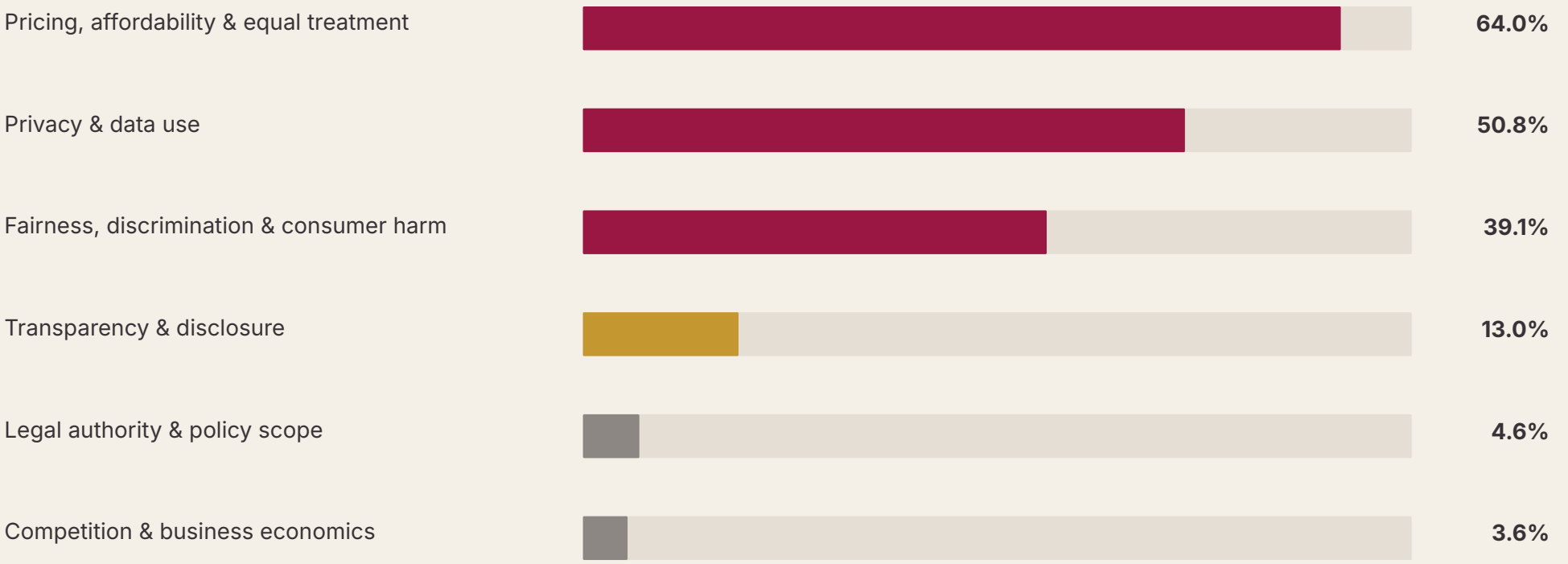
963 of 1,566 comments were classified as stronger action than the proposal itself.



Interpretation limit: these are model-coded frequencies in a self-selected analytic corpus, not estimates of representative public opinion.

The concerns went well beyond disclosure

Policy concerns were multi-label, so percentages overlap.



Only 13.0% received the transparency/disclosure label, while pricing, privacy, and fairness concerns were much more common.

The model was cheap. Responsible use still needs people.

Two numbers capture the operational trade-off.

32.6%

510 comments entered the combined governance review queue.

The queue combines the model review flag with evidence-grounding failures. It is not identical to the 511 comments classified "unclear."

~$0.69

Observed Claude model-inference cost across validation, full-corpus classification, and stability runs.

Not included: our time, data engineering, quality assurance, security, documentation, governance, or future human review.

The value is not replacing analysts; it is concentrating their attention on the cases that most need judgment.



Claude Haiku 5.5 pricing checked November 10, 2026; observed token counts saved in project outputs.

Our decision: assisted use, not autonomous reporting

We translated the class governance workshop into measurable operating rules.

Performance floor

Macro-averaged F1 score ≥ 0.80

PASS · 0.820

Below the floor, automated aggregate reporting is not permitted.

Stability target

Three-run exact agreement ≥ 90%

TRIGGERED · 86.7%

Keep human review and investigate unstable coding boundaries.

Subgroup monitoring

Error-rate ratio ≤ 1.25× best length group

TRIGGERED

Review performance gaps across comment-length groups before broader use.

Risk scoring and operating rules are auditable: action required when a metric crosses a threshold, with the control living in policy, prompt, or code.

WHAT WE LEARNED

The hard part was not calling a model.

1 **Define the decision before the labels.**

Policy intent was more useful than generic sentiment.

2 **Human coding made the model measurable.**

The frozen codebook gave us a benchmark and exposed difficult boundaries.

3 **Cheap inference did not remove governance.**

Low model cost coexisted with stability and subgroup triggers and a substantial review queue.

Generative artificial intelligence can make large public-comment records tractable - but only when coding logic, evaluation criteria, and governance rules are explicit.

